

1. Focus of the Study

Negative Polarity Items (NPIs): Words such as *ever*, *any*.

They need to occur in licensing contexts (within the scope of negation) as in (1).

Otherwise, their use is ungrammatical as in (1*).

(1/1*) a. John *hasn't/*has ever* been to Istanbul.

NPI licensing involves (long-distance) dependency between **the licenser** and **the NPI**.

Memory retrieval (Lewis & Vasishth, 2005): When an NPI is encountered, a previously encountered licenser must be retrieved from memory.

In Turkish (a head-final language) NPIs precede their licensers in unmarked (basic) word order.

Surprisal (Hale, 2005; Levy, 2008): The more predictable a licenser, the easier to resolve the dependency (entropy reduction).

DLT (Gibson, 2000): The lower the memory load when a licenser is encountered, the easier to resolve the dependency (distance/locality and structural integration cost).

The Present Study: -- tests predictions of surprisal and DLT in (long-distance) licensing of NPIs in Turkish.
-- tests retroactive (matrix verb negation) interference in Turkish NPI licensing.

2. NPI Licensing in Turkish

Syntactic constraints: -- NPIs must be licensed locally within the scope of a c-commanding negation marker

-- When they occur in embedded clauses (ECs), they can, sometimes, be long-distance licensed by the matrix verb.

-- Long-distance licensing is subject to the embedded clause (EC) structure.

Transparent domain: If EC is not marked for tense, aspect and factivity as in (2a), it forms a transparent domain which allows for long-distance (matrix verb) licensing of an embedded NPI

Opaque domain: If EC is marked for tense, agreement and factivity as in (2b), it forms an opaque domain which prevents long-distance (matrix verb) licensing

(Konfilt, 1984; 2007)

(2) a. **Transparent EC:** Ali [hiç kimse-nin geç kal-**ma**-sin-i] iste-me-di.
Ali any body-GEN late stay-NOM-3SG.POSS-ACC want-NEG-PAST
Ali did not want anybody to be late.

b. **Opaque EC:** *Ali [hiç kimse-nin Ayşe-yi sev-diğ-in-i] hatırla-ma-dı.
Ali any body-GEN Ayşe-ACC like-FN-3SG.POSS-ACC remember-NEG-PAST
Ali did not remember that anybody liked Ayşe.

Semantic factors: An EC NPI can be long-distance (matrix verb) licensed if the matrix verb is a neg(ative)-raising verb.

Neg-raising verbs: Matrix verb negation can extend its scope to EC with verbs such as *think*

(3) I *don't think* he is coming.
(i) I *do not think* he is coming.
(ii) I *think* he *is not* coming.

(Horn, 1978;1995)

❖ The present study examines if EC structure results in prediction for long-distance licensing of an EC NPI.

3. Stimuli: [Subject [Subject + NPI + embedded VP] matrix VP]

Materials: 60 sentences with 12 conditions selected in a normative study (ensuring transparent/opaque EC unbias)

Licensing: *Locally licensed* (LL), *Long-distance licensed* (LDL), *Unlicensed* (UL)

EC manipulation: *Transparent* (-*ma*) or *Opaque* (-*AcAK*) domain

Matrix verb manipulation: *Neg-raising* (NR) or *non-neg-raising* (NNR) matrix verb

(4) (Negation suffix -*ma* (subject to vowel harmony) underlined.)

NR Transparent a. LL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-ma-sin-i] muhtemelen *um-uyor-du* ...
b. LDL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-sin-i] muhtemelen *um-mu-yor-du* ...
c. UL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-sin-i] muhtemelen *um-uyor-du* ...
Opaque d. LL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-yacağ-in-i] muhtemelen *um-uyor-du* ...
e. LDL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-yacağ-in-i] muhtemelen *um-mu-yor-du* ...
f. UL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-yacağ-in-i] muhtemelen *um-uyor-du* ...
Mr. Hasan probably (did not) hope(d) that the secretary would call anyone beforehand.

NNR Transparent g. LL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-ma-sin-dan] muhtemelen *yakin-iyor-du*
h. LDL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-sin-dan] muhtemelen *yakin-mi-yor-du*...
i. UL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-sin-dan] muhtemelen *yakin-iyor-du* ...
Opaque j. LL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-yacağ-in-dan] muhtemelen *yakin-iyor-du*
k. LDL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-yacağ-in-dan] muhtemelen *yakin-mi-yor-du*
l. UL Hasan Bey [sekreter-in *hiç* kimse-yi önceden ara-**ma**-yacağ-in-dan] muhtemelen *yakin-iyor-du* ...
Mr. Hasan probably (did not) complain(ed) that the secretary would call anyone beforehand.

4. Experiment 1 – Eye-tracking-while-reading

Materials: 60 sentences as in (4), each with 12 versions.

A content-neutral word (*bence* “in my opinion”) followed matrix verb

to prevent wrap-up effects.

96 grammatical fillers (24 control fillers with/without negation marking on the embedded or matrix verb)

Participants: 72 native speakers of Turkish

(data collection in progress)

EyeLink 1000 Plus to track eye-movements

Critical regions: EC verb (6th word) and its spillover & Matrix verb (8th word) and its spillover

Procedure: Read the sentences silently &

answer comprehension questions

EyeLink 1000 Plus to track eye-movements

5. Experiment 1 - Data Analysis & Results

Data cleaning: fixations < 80 ms merged with nearby fixations (1 degree visual angle)

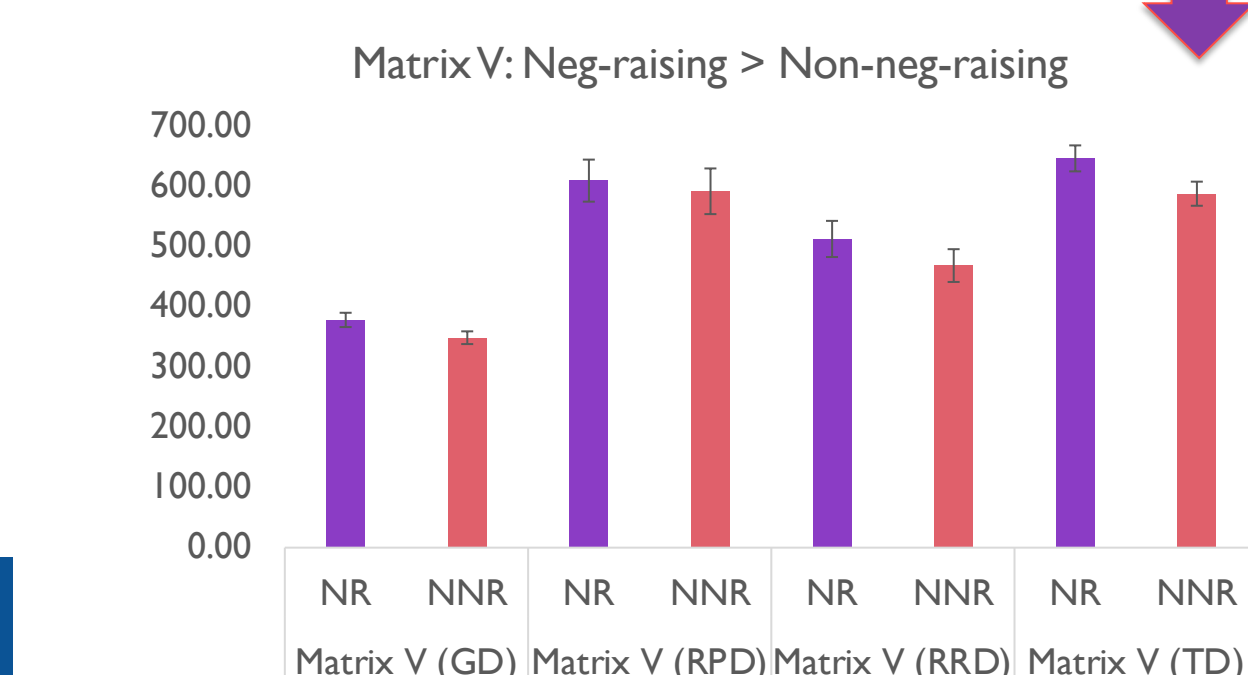
fixations < 80 ms, > 800 ms eliminated

Table 1. Means and standard errors (SE) for the embedded verb (EC-V) and its spillover (EC-S) and matrix verb (MV) for first fixation duration (FFD), gaze duration (GD), regression path duration (RPD), re-reading duration (RRD), total duration (TD) and probability of regression out (PRO)

		FFD			GD			RPD			RRD			TD			PRO		
		EC-V	EC-S	MV	EC-V	EC-S	MV	EC-V	EC-S	MV	EC-V	EC-S	MV	EC-V	EC-S	MV	EC-V	EC-S	MV
T	NR	LL	239.3	243.8	249.3	411.7	276.7	376.2	503.7	388.1	616.3	545.5	376.7	469.3	756.2	429.3	610.4	0.14	0.13
			(4.8)	(5.5)	(4.8)	(12.7)	(7.2)	(11.7)	(17.2)	(21.7)	(32.2)	(26.3)	(23.4)	(22.4)	(23.7)	(13.4)	(18.7)	(.02)	(.02)
		LDL	254.5	235.1	244.8	419.2	279.5	360.8	507.8	392.4	579	651.8	381.6	517.3	834.6	455.7	618.3	0.1	0.18
			(5)	(5)	(4.7)	(15.2)	(7.6)	(10.7)	(20.4)	(20.1)	(32.2)	(33.1)	(22.4)	(32.1)	(31.8)	(16.1)	(20.1)	(.02)	(.02)
UL		LL	254.8	238.4	251.8	412.7	278.1	389.7	509.1	403.7	666	602.1	370.9	552.8	819.7	443.8	706.8	0.12	0.19
			(3.9)	(5.6)	(5.2)	(13.4)	(7.8)	(13.6)	(19.9)	(20.1)	(41.8)	(32.7)	(24.1)	(36.6)	(30.1)	(14.9)	(24.4)	(.02)	(.02)
		NNR	244.8	240.5	252.9	426.5	271.1	344.7	549.9	443.2	548.3	662.8	355.5	469.2	858.5	446.3	597.8	0.12	0.22
			(4.8)	(4.8)	(5.3)	(13.5)	(6.8)	(10.1)	(26.3)	(25.4)	(30.3)	(37.6)	(17.9)	(27.8)	(31.7)	(14.6)	(19.7)	(.02)	(.03)
LDL		LL	250.9	242.9	242.2	448.8	278.5	342	561.8	403.2	560.4	607.4	378.5	442.4	846.8	456.5	565.6	0.12	0.2
			(4.8)	(5.2)	(5.2)	(14.0)	(9.4)	(9.9)	(22.7)	(22.2)	(31.2)	(28.2)	(22.1)	(26.7)	(25.8)	(18.3)	(20.5)	(.02)	(.03)
		UL	251.8	239.3	246.4	447.1	267.9	351.9	564.6	448.8	627.3	682.1	405.4	513.3	878.1	459.5	607.1	0.13	0.21
			(5)	(5)	(5)	(16.1)	(7.1)	(9.7)	(23.4)	(26.7)	(36.7)	(34.7)	(20.4)	(29.2)	(29.6)	(15.4)	(21.9)	(.02)	(.02)
O	NR	LL	246.3	238.1	248.7	431.2	270.6	391.6	541.3	360.1	619	561.6	407.9	522.4	783.4	457.1	654.3	0.1	0.13
			(4.6)	(5.3)	(4.8)	(14.5)	(7.1)	(12.5)	(27.7)	(20.1)	(33.5)	(32.1)	(26.7)	(34)	(27)	(16.9)	(23.2)	(.02)	(.02)
		LDL	244.3	232.3	253.4	396.1	264.8	397.9	498.6	410.7	599	618.5	376.7	501.3	780.5	438.1	668.8	0.11	0.17
			(4.3)	(4.8)	(5.7)	(11.4)	(7.8)	(12.7)	(22.9)	(30.6)	(32.9)	(30.1)	(23.1)	(25.4)	(25.5)	(18.1)	(22.9)	(.02)	(.02)
UL		LL	243.2	233.2	253	445.0	269.7	353.7	524.2	389.3	579.2	565.5	386.8	513.6	782.1	414.3	621.3	0.11	0.13
			(4.7)	(4.8)	(5)	(13.5)	(6.6)	(10.1)	(19.9)	(26.1)	(36)	(30.9)	(21.2)	(29.8)	(25.9)	(13.5)	(20.1)	(.02)	(.02)
		NNR	247.5	237.3	242.2	424.8	266.5	357.6	502.7	352.8	586.3	543.9	387.2	447	766.3	452.4	552.9	0.07	0.1
			(4.3)	(4.8)	(5.2)	(11.4)	(6.4)	(12.9)	(21.2)	(24.4)	(44.2)	(25.4)	(24.9)	(29.4)	(22.7)	(16.9)	(18.3)	(.02)	(.02)
LDL		LL	248.3	237.2	245	421.7	261.6	348.4	521.6	386.4	621.6	639.0	355.8	464.8	818.2	423.3	602.1	0.12	0.16
			(4.4)	(4.9)	(4.8)	(11.4)	(6.4)	(10.8)	(20.9)	(26.6)	(42.3)	(28.4)	(19.1)	(25.3)	(24.9)	(13.7)	(19.9)	(.02)	(.02)
		UL	247.5	237.7	249.1	435.5	269.1	348	538.2	392.2	608.1	588.6	377.8	475	830.7	441.9	602.3	0.12	0.15
			(5)	(6.2)	(4.8)	(13.6)	(8.3)	(10.1)	(23.1)	(25.2)	(42.9)	(33.1)	(23.3)	(24.9)	(29.2)	(16.6)	(19.9)	(.02)	(.02)

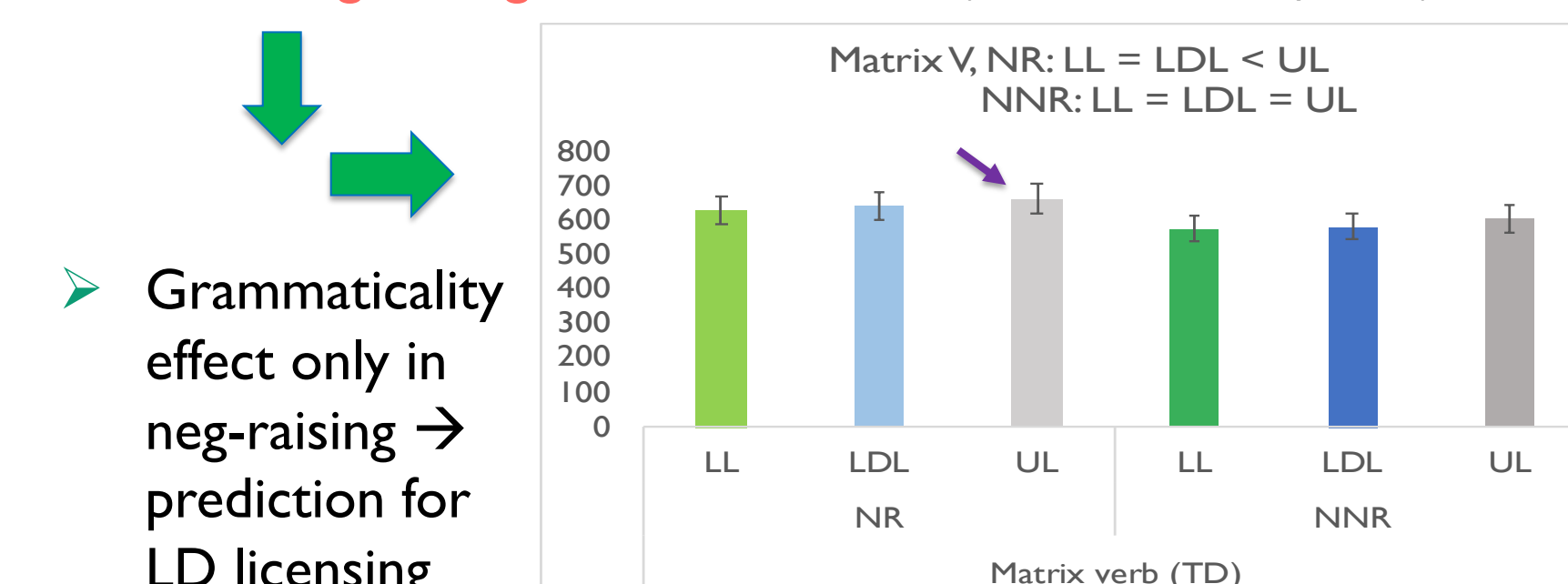
Neg-raising disadvantage in
GD, RPD, RRD and TD on the matrix verb

(θ 's ≥ 26.7 , t 's ≥ 2.53 , p 's < .05)



Neg-raising & licensing predicted TD on the matrix verb
(χ^2 's ≥ 28.36)

Neg-raising: LL = LDL < UL ($\theta = 44.5$, $t = 2.14$, $p < .05$)
Non-neg-raising: LL = LDL = UL ($\theta = 24.1$, $t = 1.28$, $p = .20$)



➤ Grammaticality effect only in neg-raising → prediction for LD licensing

6. Experiment 2 – Acceptability Judgments

Materials: Same as in Experiment 1.

26 ungrammatical & 72 grammatical fillers (proportional to critical items)

Participants: 56 native speakers of Turkish

(data collection in progress)

Procedure: Rated the sentences on a scale of 1 to 7 (acceptable) on PCLbex

Table 2. Means and standard errors (SE) for the acceptability ratings

		Transparent			Opaque		
		LL	LDL	UL	LL	LDL	UL
NR		5.13	2.84	2.14	5.33	3.16	2.15
		(.11)	(.12)	(.09)	(.11)	(.12)	(.09)
NNR		5.76	2.73	2.29	5.72	2.53	2.02
		(.10)	(.11)	(.10)	(.10)	(.11)	(.08)

➤ Effects of neg-raising and licensing in ungrammatical (NNR, O) suggests evidence for **grammaticality illusion** (retroactive intrusion)

Predictions:

Memory retrieval: Advantage for local-licensing but no specific predictions for sub-conditions for a forward-looking dependency (unless retroactive interference).

Suprisal: Prediction for long-distance licensing based on the embedded clause (EC) structure:

-- Opaque EC: grammaticality decision at the embedded verb

-- Transparent EC: No penalty for unlicensing at the EC, wait for the matrix verb.

-- LDL is easier when predicted as in neg-raising and/or transparent domain.

DLT: Advantage for local-licensing

Opaque > transparent due to tense & agreement projections (structural integration cost).

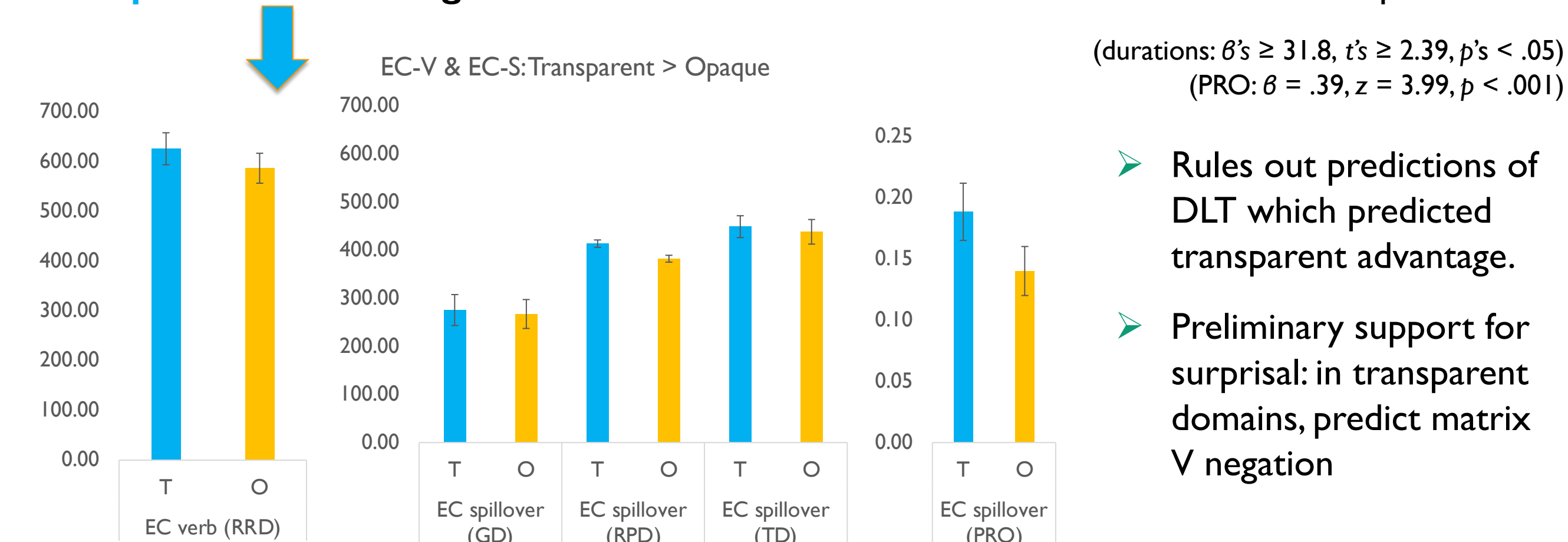
Data analysis: Mixed effects linear/logistic regression with *lmer* package

fixed factors: licensing (locally licensed, long-distance licensed, unlicensed), EC structure (transparent & opaque) and matrix verb type (neg-raising & non-neg-raising)

random factors: subjects, items

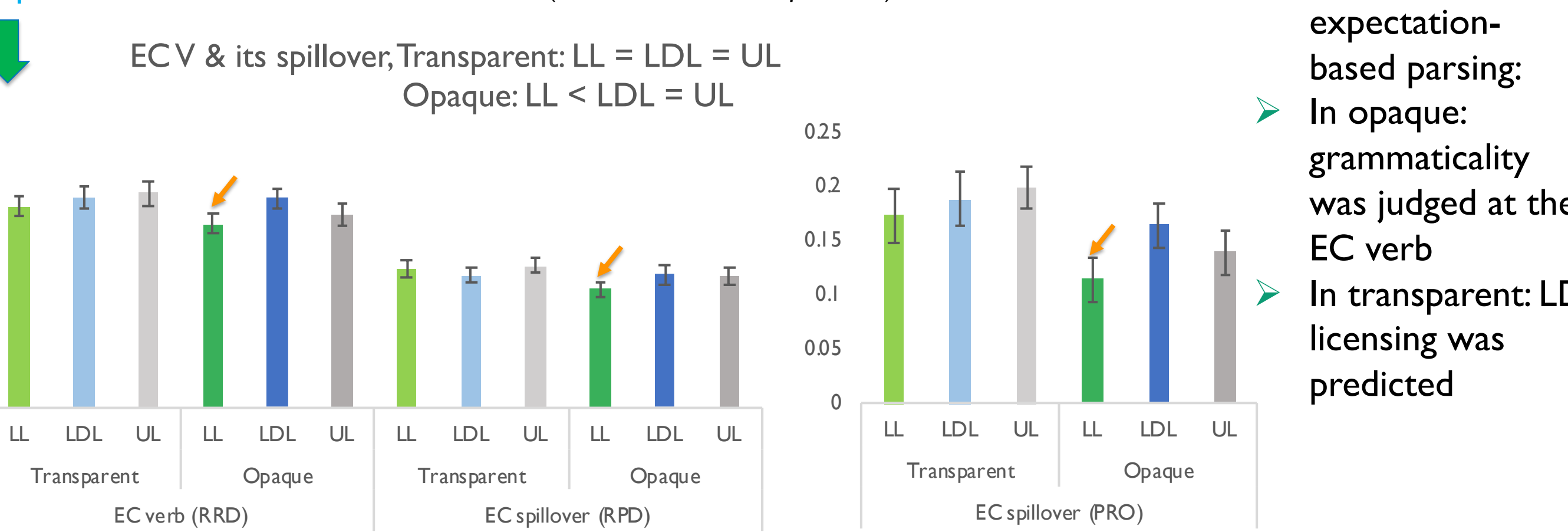
measures: six standard eye-tracking measures (see Table 1)

Transparent disadvantage in RRD on the EC verb and GD, RPD, TD and PRO on its spillover



EC structure & licensing predicted RRD on the EC verb & RPD and PRO on its spillover region
(χ^2 's ≥ 8.51)

Opaque: LL < LDL = UL (θ 's ≥ 56.4 , t 's ≥ 1.94 , p 's < .05)
Transparent: LL = LDL = UL (θ 's ≤ 29.7 , t 's ≤ 1.30 , p 's $\geq .20$)



7. Discussion

➤ Eye-tracking results support the predictions of *surprisal theory* (expectation-based parsing).

❖ Turkish speakers used EC verb morphology to predict long-distance licensing of NPIs.

❖ In ECs with opaque domain: Grammaticality was judged at the EC verb.

❖ In ECs with transparent domain: Grammaticality was not judged at the EC verb.

❖ LDL was easier with neg-raising verbs (but not with transparent domains).

➤ Support for grammaticality illusion, suggesting retroactive intrusion, in judgments.

❖ LL > LDL > UL in ungrammatical conditions (NNR & Opaque) as well as semantically (NR) or syntactically licit (Transparent) ones.

➤ Despite in different experiments, surprisal and retroactive intrusion effects for the same structure: potential for testing prediction and memory effects together?

(Campanelli et al., 2018; Parker et al., 2017).

References & Acknowledgements

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